

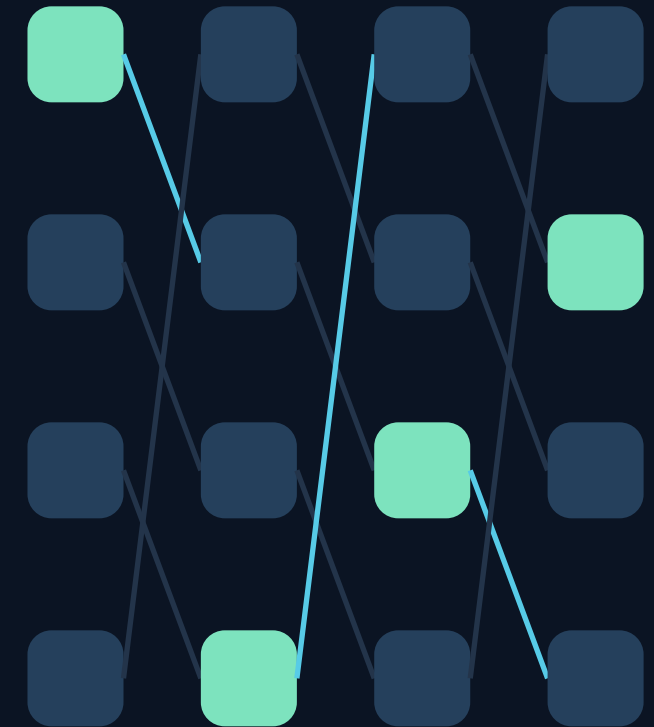
# Sleeping Machines

The next general-purpose substrate for machine intelligence.

One learner. Many forms of experience. Reusable intelligence, from representations to execution.

**Tero Keski-Valkama | Sole founder**

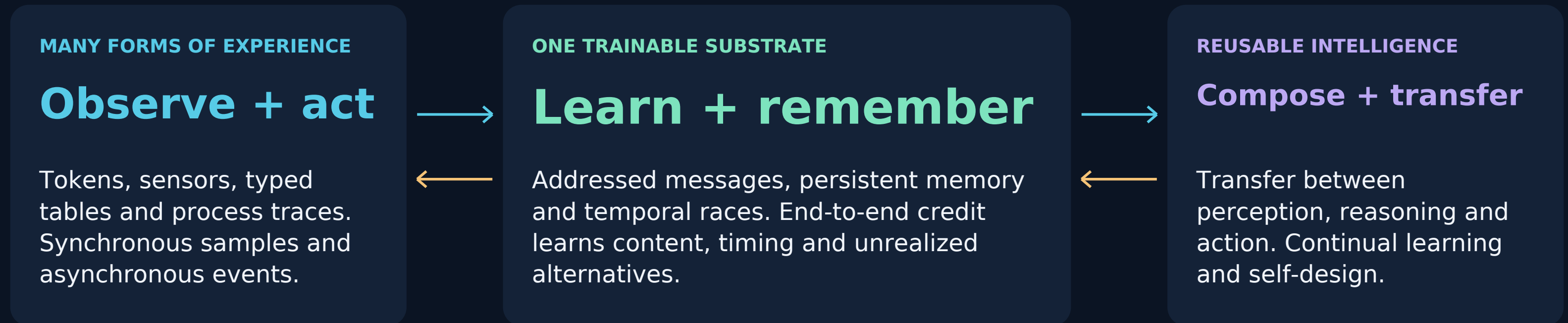
Raising €3M to scale research and engineering  
Research-stage deep tech · 4 October 2026



**Capacity beyond activity**

# One substrate for learning, representing and executing intelligence.

Different forms of experience can teach reusable computations within the same learner.



**The platform thesis: knowledge learned from one form of experience can become useful computation in another.**

Time and sleeps perform computation. Target: clockless learning and inference, capacity beyond activity, small data to large scale.

**First bridge: mixed-type tables through the temporal core. First transfer proof: joint learning, retained skills, matched work.**

# Beats the saved Transformer baselines at a fraction of the compute.

Wikipedia text (text8); same data and test set for all. Error: bits/char (bpc) or sMAPE %; lower is better.

| Comparison                                      | Their error       | Our error | Result and compute we used                                 |
|-------------------------------------------------|-------------------|-----------|------------------------------------------------------------|
| vs Transformer, 4 layers, 4 passes (10M chars)  | 1.908 bpc         | 1.888 bpc | Better · 40% of its training, 18% of its inference compute |
| vs Transformer, 2 layers, 1 pass (10M chars)    | 2.427 bpc         | 1.955 bpc | Better · 96% of its training, 16% of its inference compute |
| vs LSTM-256 (10M chars)                         | 2.171 bpc         | 1.955 bpc | Better at 88% of its inference compute (more training)     |
| vs LSTM-512, 6 passes (10M chars)               | 1.799 bpc         | 1.888 bpc | Behind by 0.09; a wider model did not close it             |
| vs best tuned LSTM (6 tuned runs)               | 1.825 bpc         | 1.888 bpc | Behind by 0.06; it used 92% of our compute                 |
| vs best tuned Transformer (4 tuned runs)        | 1.996 bpc         | 1.888 bpc | Better by 0.11 at ~equal compute (102%)                    |
| vs LSTM-512 / Transformer (90M chars)           | 1.661 / 1.604 bpc | 1.800 bpc | Behind by 0.14 / 0.20, at 1/4 / 1/8 of their compute       |
| NeuroBench chaotic forecast (official, 30 runs) | 13.37% (LSTM)     | 14.84%    | Behind by 1.5; ties ESN (14.79%); smallest model           |

**Wins vs both saved Transformers; a tuned small LSTM still leads. That gap is the next proof.**

Compute = estimated arithmetic operations. Baselines trained by us; 10 tuned runs at our two budgets complete, selected by validation. Scoreboard: evidence pack R1.

# Event-driven models aimed for parity. Ours beats small Transformers on compute.

Published event-driven and spiking language models, as reported by their authors, beside our completed result.

| Model                                       | How it computes                                                              | Reported language result                                                          |
|---------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| EGRU on SpiNNaker2 (2023)                   | Clocked recurrent net; units fire events above a threshold                   | Matches LSTMs                                                                     |
| SpikeGPT, 216M (2023)                       | Clocked binary spikes in a recurrent Transformer variant                     | “Competitive” with non-spiking models                                             |
| Loihi 2 LLM, 370M (2025); sparse SSM (2026) | Conventional models made sparse/quantized; the chip supplies asynchrony      | No accuracy loss / comparable to dense                                            |
| Timing-coded LLM, 1.5B (2026)               | Spike times carry values; closest to our races                               | “Clear gap” in language perplexity                                                |
| Sleeping Machines (ours)                    | Learned arrival times choose which memories update; unchosen paths learn too | Beats two small Transformers at 40% of training compute; tuned LSTMs lead by 0.06 |

**To our knowledge, the first event-timed model to beat Transformer baselines on language with less compute, so far at small scale.**

Different datasets and scales; not a ranking. Ours: text8, 10M characters, small models, single runs, our baselines; LSTM-512 still ahead.

# Intelligence must compose across experience and execution.

A common substrate connects what a model learns with how and where it computes.

## REPRESENTATION

### Meaningful inputs

Tokens, sensor events and typed comparisons enter through appropriate interfaces and act on shared persistent state.

## LEARNING

### Reusable skills

End-to-end credit connects perception, memory, reasoning and action. Joint training tests transfer and retention.

## EXECUTION

### Adaptive work

Learn timing and selective activity; recruit useful capacity for the task and the resources available.

**Small-data tables and large-data language are complementary tests of the same platform thesis.**

# One model family across today's architectural landscape.

Our approach combines strengths of today's main model types.

## Recurrent networks

Carry a running memory forward.  
We add addressed memory and timing.

## Transformers

Every input attends to every other.  
We keep that where it pays off.

## Mixture of experts

Activate part of a large model.  
We add private memory and timing.

## SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One interface for content, timing, memory address and state, from single units to complete systems.  
Target: computation through timing, sparse memory updates, small messages, and learning from unchosen paths.

## Statistics / retrieval

Counts and lookups of past data.  
Combined with learned features.

## Adaptive memory

Models that adapt while in use.  
With local, causal updates.

## Event-driven systems

Event-driven, neuromorphic systems.  
We add trainable messages.

These are design relationships, not benchmark claims. Each combination must still be built and measured.

# Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

## 01 / UNITS

### Local programs

Each unit chooses what to remember, what message to send and when. Timing itself performs computation.



## 02 / LAYERS

### Adaptive work

A layer can activate one message or many: dense where needed, lean elsewhere.



## 03 / COMPOSITIONS

### Shared evidence

Learned features, counts and stored history can all inform the same prediction.



## 04 / SYSTEMS

### Same learner

The same learning rule works in batches or on a live stream, including learning after deployment.

-----  
**Counterfactual credit: learn from selected and unrealized routes**

**No global clock is required. Combined models, learning-after-deployment gains and chip energy are still milestones.**

# Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

## 01 / ENCODE

### Signal + delay

A learned function produces a message and its arrival time from input and local memory.



## 02 / SELECT

### Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.



## 03 / REMEMBER

### Local state

Update the selected memory and propagate a new message into deeper computation.



## 04 / LEARN

### Alternatives

Use errors to improve message content and timing, including paths that were not selected.

-----  
**Counterfactual credit: learn from selected and unrealized routes**

**Implemented and trained today in ordinary software (CPU); no special hardware needed to run it.**

# One temporal family learns across three data structures.

Different interfaces; addressed memory, temporal computation and counterfactual route credit.

## ANONYMOUS INTERLEAVED LOGS

**0.592 vs 0.559 AUROC**

FAS at 256 events: all three native seeds beat six saved generic controls. At 512 events: mean 0.737 vs 0.727. Replicated v1 quality wins; sealed v2 follows.

## MIXED TYPES, SMALL DATA

**100% on 256 DEV rows**

64 FIT rows; 6,370 parameters. Typed comparisons feed the integrated temporal core. Category/order/missingness and alternative-write checks pass. Synthetic interaction; fixed predicates.

## TOKEN LANGUAGE, USEFUL MEMORY

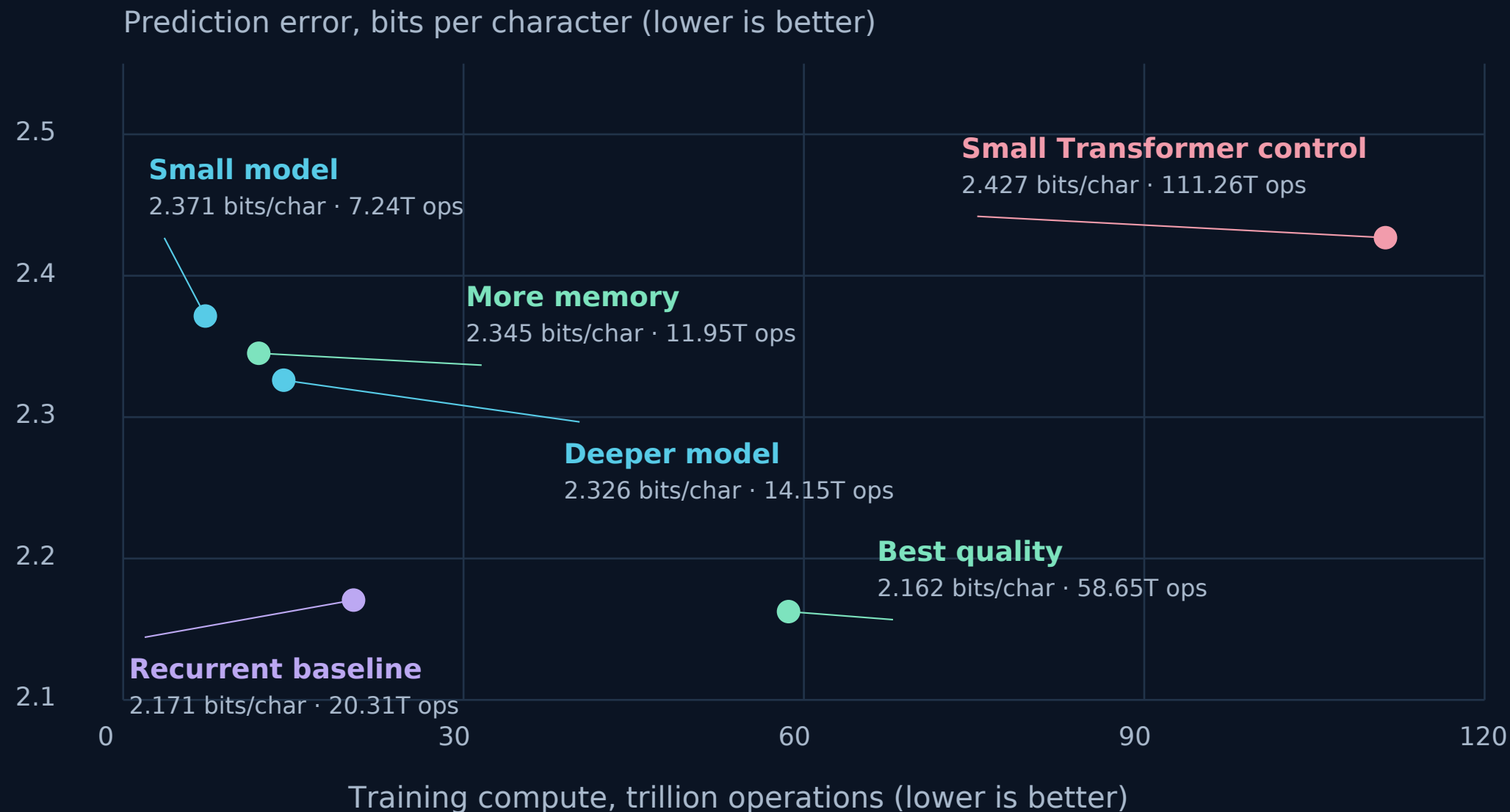
**7.2515 / 7.2636 NLL**

Two seeds at 1M TRAIN tokens. Erasing persistent memory raises loss in both. Fixed-width quality improves with data. DEV evidence; public Transformer comparison follows.

**A common construction across data forms. Next: shared skills, real tables and the reserved larger language point.**

# Single pass: error versus training compute.

Each model sees the 10M training characters once. Lower and further left is better. Slide 2 adds multi-pass runs.



## BEST SINGLE PASS

# 2.162 bpc

Edges LSTM-256 (2.171), with 2.9× its training compute.

## TRAINING COMPUTE

# 15.4× less

Small model vs 2-layer Transformer: less compute and lower error.

Baselines trained by us on the same data and test set. Compute is estimated arithmetic (trillion operations), not measured energy or cost. Single runs.

# Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

## WHO WOULD EVALUATE IT

### Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

## WHAT WE WOULD SUPPLY

### Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

## WHY THEY WOULD ADOPT

### Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

**Commercial gate: target  $\geq 20\%$  full-system cost reduction at matched quality on one selected workload.**

# Software proves the advantage. Hardware multiplies it.

One research program, with staged routes to deployment.

## 01 / MODELS + RUNTIME

### First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

## 02 / MEMORY-LOCAL ACCELERATORS

### 1.7–5.8× less data

Modeled per character: 1.7× fewer memory bytes than the best tuned Transformer and 5.8× fewer than the saved one, both beaten. 10× more memory adds 5% traffic. Tuned LSTMs: parity.

## 03 / CLOCKLESS SILICON

### Work only on events

No global clock: idle parts draw almost no power, answers arrive as soon as they are ready, and cool chips allow dense 3D stacking. Also the missing model for neuromorphic chips.

**The €3M round funds scale and a hardware cost model of the trained network; silicon follows measured proof.**

# A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

## AMD

### Model + platform

Its ~\$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

## INTEL

### Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

## GOOGLE

### Full-stack design

TPU model-hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

**Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.**

# Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

| Per-customer assumption | Calculation                                        | Annual amount |
|-------------------------|----------------------------------------------------|---------------|
| Eligible workload cost  | Existing deployment cost we can actually influence | €1.00M        |
| Gross savings           | 20% complete-cost reduction at matched quality     | €200k         |
| Runtime license         | 20% of verified gross savings                      | €40k          |

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

**Validate pricing through matched quality, reliable operation and measured customer benefit.**

# AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

## Tero Keski-Valkama

### Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

**2012-2018**

**Cybercom · Software Architect**

Historical role in public professional profile.

**2018-2022**

**HERE · Lead ML & AI Engineer**

Self-reported research and applied ML leadership.

**2022-2025**

**Alloy.ai → Kaiko.ai**

Staff engineering; multimodal foundation-model work.

**2023 publication**

**EP4148389A2 · Named inventor**

Maps from geospatial observations; applicant HERE.

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

# Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

## MODEL + LEARNING

### Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

## RUNTIME + HARDWARE

### Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

## RIGHTS + PATENTS

### Company assets

Secure company ownership. Assess technical inventions for selective patent filings; protect confidential implementation know-how.

**Funding builds company-owned IP: ownership review, selective patent filings and confidential know-how.**

# Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

## 01 / QUALITY AT SCALE

### Replicate

At least three runs per result, fixed held-out tests and tuned baselines. Compare quality and total compute; publish failures too.

## 02 / COMPLETE ECONOMICS

### Measure

Show the fast sparse runtime gives the trained model's answers. Measure serving cost, latency, memory and energy on one workload.

## 03 / ADOPTION

### Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

**If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.**

# The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

## MONTHS 0-3

### Team + proofs

Hire the initial team.  
Resolve IP ownership and review priority filings.  
Verify the trained sparse runtime; repeat larger-data results.

## MONTHS 3-6

### One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

## MONTHS 6-12

### External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

## MONTHS 12-18

### Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

**Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.**

# €3M funds an 18-month research and engineering program.

Team, staged compute, hardware feasibility and IP protection, including selective patent filings.



- Team: research + systems €1.35M | 45.0%
- Compute + independent replication €0.90M | 30.0%
- Hardware feasibility + measurement €0.25M | 8.3%
- IP, legal, operations €0.20M | 6.7%
- Contingency reserve €0.30M | 10.0%

**€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.**  
Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.

# Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

## CAPITAL TO RAISE

**€3M**

18 months of research, systems engineering and staged validation.

## PROPOSED PRE-MONEY

**€50M**

A premium research-stage proposal for the model/runtime/substrate opportunity.

## INITIAL OWNERSHIP

**5.66%**

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future chip architecture that already beats Transformer baselines at a fraction of the compute. A confirmed 90M-character win and a NeuroBench first place would justify a higher next price (the €100M scenario in the appendix).

**Valuation sensitivities and conditional exit scenarios are in the appendix.**

# Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

## €3 million

Already beating Transformer baselines on compute.  
This round tests whether the advantage scales.

- 01** Replicated language quality and learning mechanisms.
- 02** A deployable runtime with measured system economics.
- 03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement

# The same model improves when unchosen paths receive credit.

Controlled small-model comparison; held-out error and estimated complete training work.

## TEST BITS / CHARACTER

**2.506 → 2.371**

Timing-only credit versus added local alternative-value route credit.

## ESTIMATED FITTING WORK

**+0.30%**

Hard forward selection and parameter count retained.

## CHARACTER PERPLEXITY

**−8.9%**

Derived from the BPC difference; one controlled seed.

## Same small model; hard selection retained

A deeper variant also improves by 0.130 bits per character. Both are single-seed comparisons; numerical details and model definitions follow.

**A falsifiable architectural mechanism, with a small counted arithmetic increment.**

# More available state. The same eight selected writes.

Small model with alternative credit: available memory doubles; selected updates stay fixed.

## Pool 2



16 slots · 512 value scalars  
8 writes · 16 keys scored  
2.371 bpc

## Pool 4



32 slots · 1,024 value scalars  
8 writes · 32 keys scored  
2.345 bpc

## WHAT SCALES

# 8 writes

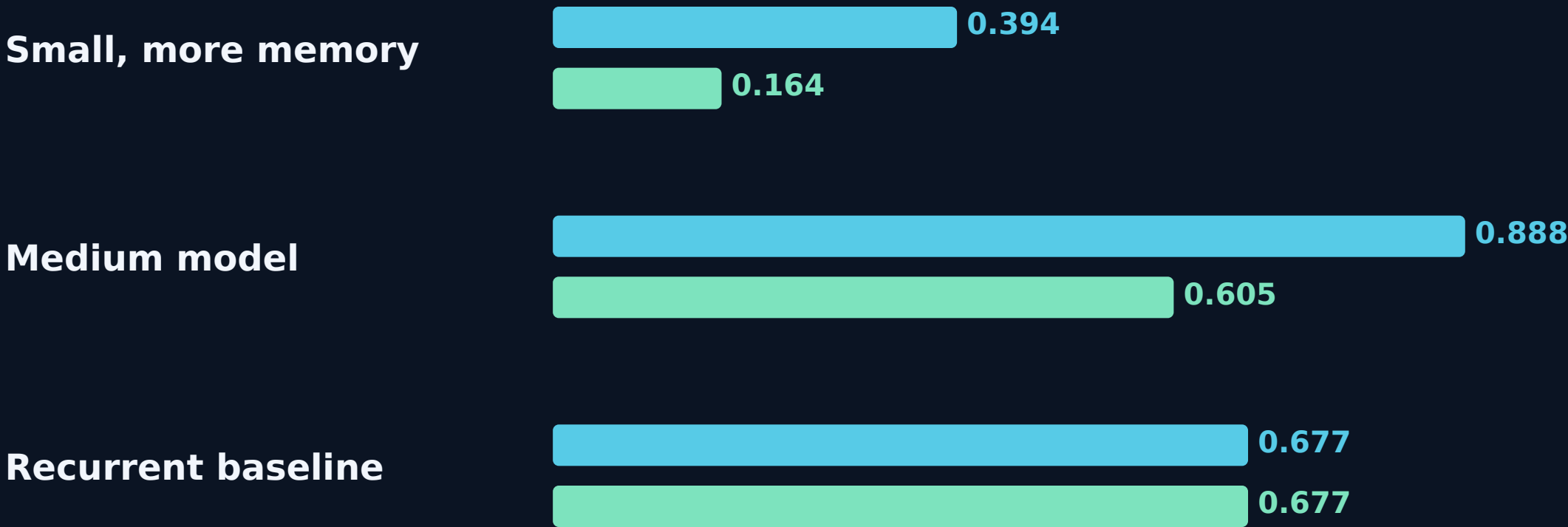
Winner-only arithmetic +0.66%.  
Fitting work +65.0%. Keys and  
value storage double.

Quality improves by 0.026 bpc; total capacity and learning costs remain explicitly charged.

# The sparse path is implemented; system proof is next.

Inference arithmetic per predicted character. Green bars: our implemented sparse runtime, which computes only the selected paths.

Inference compute per character, million operations (lower is better)



**STATUS**

Every memory address is still scored. Checking that the sparse runtime reproduces the trained model exactly is pending, as are measured traffic, latency and energy.

Cyan: full computation (all paths) Green: sparse runtime, only the selected paths (the baseline has no sparse mode)

Arithmetic estimates, not measured service cost. Our evaluation also pays about two extra positions of warm-up per predicted character; the LSTM evaluates differently.

# An adaptation signal exists. Its work is accounted for.

Predict-before-update experiment on a new text stream; separate from the latest model family.

## PREDICT BEFORE UPDATE

**3.191 → 3.096**

8,191 new development targets; same persistent state, frozen versus full-backbone adaptation.

## COSTED LEARNING

**10.7× work**

Total online processing work includes credit, backward and optimizer. Retention under drift and native asynchronous learning remain open.

**Online-learning upside: persistent adaptive systems. Current result: one costed pilot.**

# One substrate can support new communication and learning patterns.

Application hypotheses with first proofs; no codec, embodied-transfer or AGI result yet. The first row needs no special hardware.

| Opportunity                                     | Architectural fit                                                                                | First discriminating proof                                                                                                             |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Event-log anomaly detection (existing hardware) | Built-in biases: computational time, first-arrival races, separate state per concurrent process. | FAS v1 is solved by a simple route rule (AUROC 0.755 at 256 events vs our 0.600); proof needs a harder v2 with ambiguous interleaving. |
| Irregular inputs                                | Addressed events retain time and update persistent programs. Dense regions remain available.     | Same causal data/queries; compare timestamp-aware controls and burst/idle resource use.                                                |
| Asynchronous codecs                             | Learn what, where and when to send; decoder retains predictive state.                            | Actual bytes versus distortion/task quality, sender/receiver work and causal latency.                                                  |
| Memory meta-learning                            | Slow rules learn fast writes, retention, retrieval and scheduling from later utility.            | Held-out delayed-use tasks; complete useful-return credit against immediate utility.                                                   |
| Embodied cognition                              | Shared sensing/action/language paths can exchange learned physical abstractions.                 | Motor-to-cognitive and reverse transfer measured separately; matched data/work and isolated-path controls.                             |

# The contribution is the complete construction.

Established alternatives already address parts of this problem; they remain essential controls.

| Approach             | Established strength                                                          | Our comparison obligation                                               |
|----------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Transformer / MoE    | Strong quality; MoE separates available parameters from selected expert work. | Match quality/data and charge routing, memory traffic and training.     |
| RNN / selective SSM  | Persistent state and efficient sequence processing.                           | Test whether temporal races add useful quality or allocation.           |
| Neuromorphic systems | Event-driven hardware and local learning precedents.                          | Prove expressive model quality, mapping and measured energy.            |
| Sleeping Machines    | Joint timing, sparse memory, deep representation and alternative credit.      | Scale integrated mechanisms without replacing them with dense carriers. |

Sparsity, recurrence and on-chip learning alone are not the novelty claim.

# What clockless, memory-local silicon can buy.

Physical mechanisms, their basis and what must be measured. No chip exists yet.

| Benefit           | Mechanism                                                                                                                 | Proof still needed                                          |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Energy            | Cost model: 1.7–5.8× fewer bytes/char than the Transformers we beat; tuned LSTMs at parity; no clock tree.                | Quality at large pools; chip-level measurement.             |
| Latency           | Average-case completion; events propagate when ready; races resolve at first arrival.                                     | Event-trace latency on FPGA semantics prototype.            |
| Cooling & density | Low average power relieves the power-density limit: memory-on-logic 3D stacking, chiplets without a global timing domain. | Activity and peak-burst profiles on real streams.           |
| Communication     | Small messages sent only when events occur, not full vectors every clock: wire energy and wiring area track event rate.   | Measured message rates and header overhead.                 |
| Local memory      | Addressed persistent state maps to banks beside compute; non-volatile memory makes dormant capacity near-free.            | Locality and reuse-distance profile of trained models.      |
| On-chip learning  | Route credit and delay/gate updates are local quantities.                                                                 | Event-local learning rule matching current trainer quality. |
| Also              | Near-threshold voltage, native event sensors, low EMI, defect tolerance.                                                  | Hardware partner evaluation.                                |

**The neuromorphic hardware class exists; it lacks a competitive trainable model. That is the opening.**

# AMD's Silo AI deal supports the strategic logic.

Announced approximately \$665M all-cash acquisition in 2024; a mature-company precedent.

## WHAT AMD ACQUIRED

### Team + stack

AI scientists and engineers, enterprise solutions, model development and software capability on AMD hardware.

## THE LUMI CONNECTION

### Poro / Viking

AMD highlighted multilingual Poro and Viking models trained on AMD-powered LUMI, with university and supercomputer collaborators.

## OUR LESSON

### Prove leverage

Build a capability that improves a platform's economics and adoption. One small research project is not a direct pricing comparable.

**Efficient models can matter to a chip ecosystem; training on its chips is one part of the value case.**

# The constraints reach from power grids to robot batteries.

Economic context for efficient intelligence; these figures are not our revenue market.

## DATACENTERS

**485 → 950**

TWh/year, 2025 → 2030 central IEA projection. Power and data movement make system efficiency economically important.

## MOBILE

**5.8B**

Unique mobile subscribers in GSMA's 2026 report. On-device inference creates a prospective route to broad distribution.

## ROBOTICS

**5M**

Industrial robots operating in 2025. Sparse sensing, persistent state and adaptation are prospective use cases.

**Select one measurable initial workload; expand only after quality and system gains hold.**

# Successful platform adoption can support very large outcomes.

Conditional commercial arithmetic illustrates the scale required; no forecast or market multiple claim.

| Successful outcome assumption    | Value capture assumption                                       | Equity-value scenario |
|----------------------------------|----------------------------------------------------------------|-----------------------|
| €10B/year eligible compute spend | 20% savings × 20% capture = €400M/year; 25× revenue assumption | €10B                  |
| €50B/year eligible compute spend | 20% savings × 25% capture = €2.5B/year; 20× revenue assumption | €50B                  |

A cross-platform substrate could participate in cloud inference, persistent agents and embodied systems. Reaching these outcomes requires sustained quality, scale, margins and broad adoption.

**The upside is large if the platform works and wins adoption. Its probability remains unmeasured.**

# The completed comparisons, using consistent resource units.

Frozen 10M-character, one-pass evidence. Model definitions are on the next slide.

| Model                      | Error, bpc ↓ | Parameters (K) | Training, TFLOPs ↓ | Train MFLOPs/char ↓ | Full run MFLOPs/char ↓ | Sparse run MFLOPs/char ↓ |
|----------------------------|--------------|----------------|--------------------|---------------------|------------------------|--------------------------|
| Small: timing-only credit  | 2.506        | 108.9          | 7.22               | 0.722               | 0.240                  | 0.163                    |
| Small: alternative credit  | 2.371        | 108.9          | 7.24               | 0.725               | 0.240                  | 0.163                    |
| Small: more memory         | 2.345        | 177.0          | 11.95              | 1.196               | 0.394                  | 0.164                    |
| Deeper: alternative credit | 2.326        | 210.0          | 14.15              | 1.416               | 0.463                  | 0.310                    |
| Medium: alternative credit | 2.183        | 422.5          | 26.79              | 2.680               | 0.888                  | 0.605                    |
| Medium: more memory        | 2.179        | 689.8          | 44.07              | 4.410               | 1.458                  | 0.607                    |
| Best quality: wider model  | 2.162        | 940.9          | 58.65              | 5.868               | 1.947                  | 1.324                    |
| Recurrent baseline (LSTM)  | 2.171        | 338.4          | 20.31              | 2.032               | 0.677                  | 0.677                    |
| Small Transformer control  | 2.427        | 1658.9         | 111.26             | 11.133              | 3.710                  | 3.710                    |

↓ = lower is better. bpc = bits per character of prediction error. TFLOPs/MFLOPs = trillion/million arithmetic operations (estimates). Train = per training character; run = per predicted character. Full run computes every path; sparse run only the selected ones (exact-match check on trained models pending). Full 15-row CSV retains older models.

# Plain-language labels map to reproducible model records.

Message width counts scalars; available memory counts local state slots.

| Deck label          | Width / layers   | Memory slots               | Record configuration                 |
|---------------------|------------------|----------------------------|--------------------------------------|
| Small model         | 32 / 4           | 16                         | p32 / d4 / pool2                     |
| Small: more memory  | 32 / 4           | 32                         | p32 / d4 / pool4                     |
| Deeper model        | 32 / 8           | 32                         | p32 / d8 / pool2 / skip2             |
| Medium model        | 64 / 4           | 16                         | p64 / d4 / pool2                     |
| Medium: more memory | 64 / 4           | 32                         | p64 / d4 / pool4                     |
| Best quality        | 96 / 4           | 16                         | p96 / d4 / pool2                     |
| Saved controls      | 256 hidden width | Conventional architectures | LSTM-256; Transformer-256 × 2 layers |

# The comparison boundaries are part of the evidence.

Maintain the original records and disclose what a result actually measures.

| Boundary          | Current completed evidence                                                                                                      | Remaining work                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Data / evaluation | Frozen table: 10M training characters. Report: completed 90M run, 1.8573 bpc on test characters 95M-96M, 256-character windows. | Align modern controls, target coverage and fit work; full official 5M test remains open. |
| Learning / seeds  | 10M ledger: 1,220 updates, lr .004 cosine; one seed; state reset per fit segment.                                               | Replication, robust recipe, depth/horizon and matched larger-data comparisons.           |
| Sparse execution  | All keys scored; selected values only in winner evaluator; small random FP64 equality fixtures.                                 | Actual trained FP32 winner/state/cache equality and held-out rescore.                    |
| System economics  | Arithmetic traces/formulas; persistent local state within windows.                                                              | Setup, warmup, traffic, memory, throughput, latency, joules and total cost.              |
| Adaptive workload | Fixed scored queries; declare late/missing readout, feedback and fallback.                                                      | Charge attempted work; report coverage separately if abstention is the task.             |

# Keep the failures beside the positive mechanism results.

A disciplined research program should revise interpretations when evidence changes.

| Finding                                      | Interpretation                                                                     | Response                                                                 |
|----------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Uncredited D8 pool expansion worsens quality | More capacity needs learnable selection; it is not automatically useful.           | Retain negative pair; test credited expansion with matched depth/seed.   |
| Write-credit variants regress or diverge     | A particular local estimator is unstable; not all route credit is equivalent.      | Withdraw variants; preserve original result and derive corrected credit. |
| Best p96 costs more than LSTM                | Small quality lead is not comparable-quality resource superiority.                 | Tune and scale; require complete quality-cost frontier.                  |
| Event-vision screening is mixed              | Small coarsening gains do not clear every full-screen seed or strong RBF controls. | Avoid cross-domain supremacy claim; keep modality-specific gates.        |

# Reverse-underwriting shows the assumptions behind a price.

This arithmetic is a sensitivity tool; the research does not estimate platform-success probabilities.

## BULLISH PRICED ROUND

**€50M**

€3M raise → €53M post-money;  
5.66% new-investor ownership.

## REVERSE UNDERWRITING

**6.74%**

Required platform-success  
probability in the stated €10B exit  
scenario.

## STRETCH SCENARIO

**€100M**

€100M pre-money requires 13.49%  
in the same model; 2.91%  
ownership.

$$V = p \times \text{exit equity} \times \text{future retention} / (1 + \text{discount rate})^{\text{years}}$$

Assumptions: €10B exit; 30% stake retention; 10 years; 15% discount; failure value zero.

**Required odds are reverse-calculated conditions. The benchmarks do not estimate those probabilities.**

# The price depends on dilution, exit scale and time.

Required platform-success odds at 10 years and 15% discount; zero failure value.

| Exit equity value | Investor stake kept after dilution | Success odds needed at €50M | Success odds needed at €100M |
|-------------------|------------------------------------|-----------------------------|------------------------------|
| €3B               | 10%                                | 67.43%                      | 134.9% (infeasible)          |
| €3B               | 30%                                | 22.48%                      | 44.95%                       |
| €3B               | 50%                                | 13.49%                      | 26.97%                       |
| €10B              | 10%                                | 20.23%                      | 40.46%                       |
| €10B              | 30%                                | 6.74%                       | 13.49%                       |
| €10B              | 50%                                | 4.05%                       | 8.09%                        |
| €30B              | 10%                                | 6.74%                       | 13.49%                       |
| €30B              | 30%                                | 2.25%                       | 4.50%                        |
| €30B              | 50%                                | 1.35%                       | 2.70%                        |

Sensitivity assumptions only. Probabilities above 100% are infeasible in this model. Preferences, option pools and intermediate financing cash flows are omitted.

# A private evidence pack and primary external context.

Source labels in slide footers are clickable; companion files must travel with the PDF.

**R1 Private completed language evidence ledger (3 October 2026)**

Saved native fits and E64 controls. Original JSON records and SHA256 hashes are supplied in the private diligence pack.

**R2 Private online-learning experiment, 8,191 targets**

Predict-before-update pilot: quality and complete processing-operation estimates; one checkpoint, stream and learning rate.

**R3 Architecture, numerical contracts and report**

Private technical report: whole-family definition, landscape, composition and scoped evidence. Modern matched controls, trained sparse parity and measured hardware energy remain open.

**M1 IEA — Key Questions on Energy and AI, executive summary**

485 TWh datacenter electricity in 2025; central projection 950 TWh in 2030. Sector context, not addressable revenue.

**M2 GSMA — The Mobile Economy 2026**

5.8 billion unique mobile subscribers and 8.8 billion connections. Subscribers are not annual device sales or customers.

**M3 IFR — World Robotics 2026 release, 24 September 2026**

5 million operating industrial robots in 2025 and more than 600,000 annual installations; no adoption by this project.

**T2 Fedus et al. — Switch Transformers**

Capacity beyond selected computation has established MoE precedent; sparsity alone is not a novelty claim.

# Public precedents establish relevance, not endorsement.

Historical roles are self-reported; inventorship is checked against the primary published document.

**S1**      **AMD — Silo AI acquisition completion, 12 August 2024**

Announced approximately \$665M all-cash acquisition; Poro and Viking on AMD. Mature-team strategic precedent, not a pre-seed valuation comparable.

**S2**      **AMD — Acquisition announcement and LUMI collaboration, 10 July 2024**

AMD highlighted enterprise solutions, expertise, software and Poro/Viking training on LUMI using AMD hardware; no single exclusive acquisition cause established.

**S3**      **Liquid AI — \$250M Series A announcement, 13 December 2024**

Financing announcement and AMD collaboration on efficient models; funding amount is not a disclosed company valuation.

**S4**      **AMD Ventures — investment focus**

AI models, infrastructure and silicon investment focus; published Series A-C focus means this project may be earlier than typical admission.

**S5**      **Intel — Hala Point research system, 17 April 2024**

Loihi 2 neuromorphic research prototype, event computing and prospective continuous learning. No Intel interest in this project claimed.

**S6**      **Google — Eighth-generation TPU architecture announcement**

TPU 8t/8i, model-hardware co-design and sparse MoE infrastructure. Existing internal capability creates both potential fit and competition.

**F2**      **European Patent Office — published application EP4148389A2**

Primary published document names Tero Juhani Keski-Valkama as inventor and HERE Global B.V. as applicant. Experience evidence, not Sleeping Machines-owned IP or proof of a grant.

# Prior event-driven language models.

Summaries of each paper's own claims; datasets, scales and compute conventions differ from ours.

- P1**     **Nazeer et al., Language Modeling on a SpiNNaker 2 Neuromorphic Chip (2023)**  
Event-based GRU on a neuromorphic chip; authors: first neuromorphic language model to match LSTMs.
- P2**     **Zhu et al., SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks (TMLR 2024)**  
216M-parameter spiking language model; authors: competitive with non-spiking models; fewer operations projected on neuromorphic hardware.
- P3**     **Abreu et al., Neuromorphic Principles for Efficient Large Language Models on Intel Loihi 2 (2025)**  
370M MatMul-free model quantized for Loihi 2; no accuracy loss versus its unquantized version.
- P4**     **Richter et al., Event-Driven Language Models with Sparse Neural Activity for Neuromorphic Hardware (2026)**  
Thresholded activations in quantized linear-attention models; comparable to dense models with up to 4x fewer effective operations.
- P5**     **Zhao et al., Large Language Models with At Most One Spike per Neuron (2026)**  
Time-to-first-spike coding on GPT-2/BERT, up to 1.5B parameters; authors report a clear perplexity gap on language modeling.

# The language experiments establish three useful facts.

Wikipedia text (text8), 10M characters of training. Prediction error in bits per character: lower is better.

## IT LEARNS REAL TEXT

# 1.888 bits/char

Better than the 4-layer Transformer baseline (1.908) with 40% of its training compute. The larger LSTM (1.799) is still ahead.

## TRAINING UNCHOSEN PATHS HELPS

# −0.135 bits/char

Giving learning signal to the paths the model did not take cuts the error of the same small model, for 0.3% more training compute.

## MORE MEMORY, SAME RUNNING WORK

# 16 → 32 slots

Doubling memory slots lowers the error while each input still updates only 8 slots. Training cost rises; running cost barely changes.

**Single runs on one dataset. The next proof is repeated runs plus a measured cost advantage.**